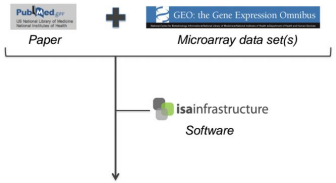
Big Data to Knowledge(BD2K)

Best Practices for Biomedical Research Data Management

Module 3

Contextual Details

Needed to make data meaningful



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Learning Objectives

1. Explain what metadata is
2. Explain why metadata is important
3. Identify and describe best practices related to metadata creation
4. Describe some common metadata elements
5. Plan an approach to creating metadata for a project

Big Data vs Little Data

- Big Data is only possible with Little Data
- Having the right data is better than no data
- Data must include provenance, metadata, and documentation

What is Metadata?

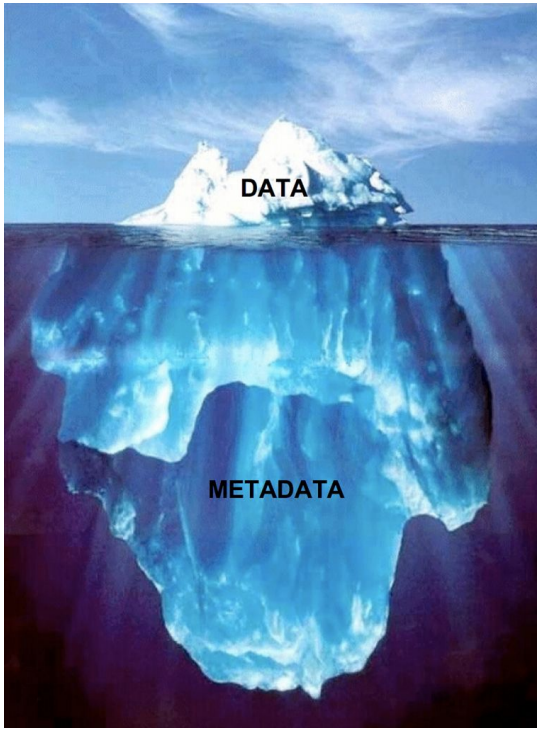


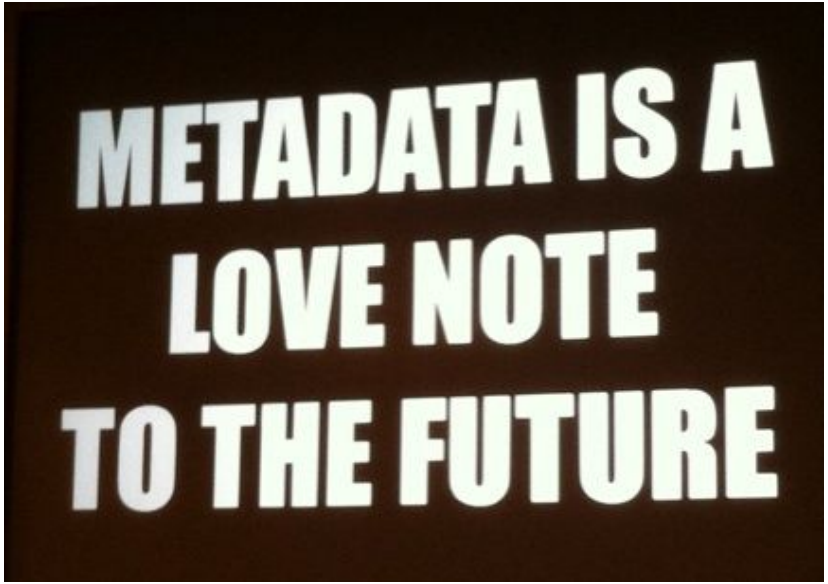
Image from Soumitri Varadarajan blog. Iceberg image
© Ralph A. Clevenger. Flickr photo

"Metadata is structured information that describes, explains, locates, or otherwise makes it easier to retrieve, use, or manage an information resource. Metadata is often called data about data or information about information"

(NISO, Understanding Metadata 2004;1)

You Must Have Metadata to:

- find data
- use the data
- help other professionals to find and use data
- use your own data in the future



Basic Types of Metadata

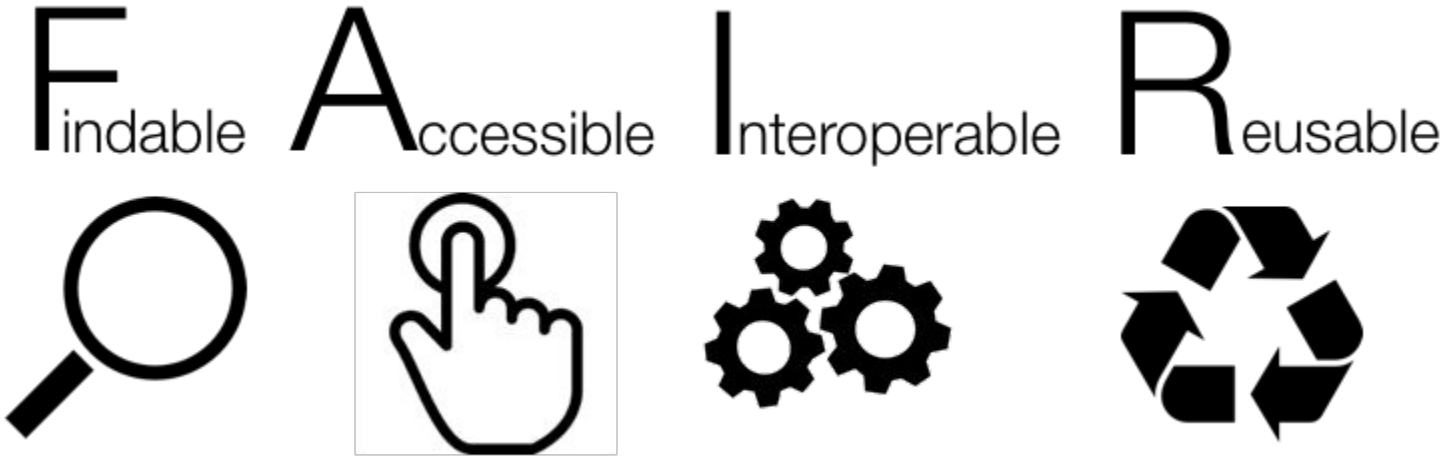
- **Descriptive metadata** describes the object or data and gives the basic facts

- **Structural metadata** describes the structure of an object including its components and how they are related

- **Administrative metadata** includes information about the management of the object

All three should give you a (mostly) complete picture of the object's content, context and structure.

How Metadata Facilitates Discoverability and Reuse



http://insideuniprot.blogspot.com/2016/11/being-fair-at-uniprot.html

Metadata Standards

There are many different metadata standards and specifications, discipline, or domain, specific.

- Dublin Core for biological data
- Darwin Core for resource description
- Genome Metadata for single genomes

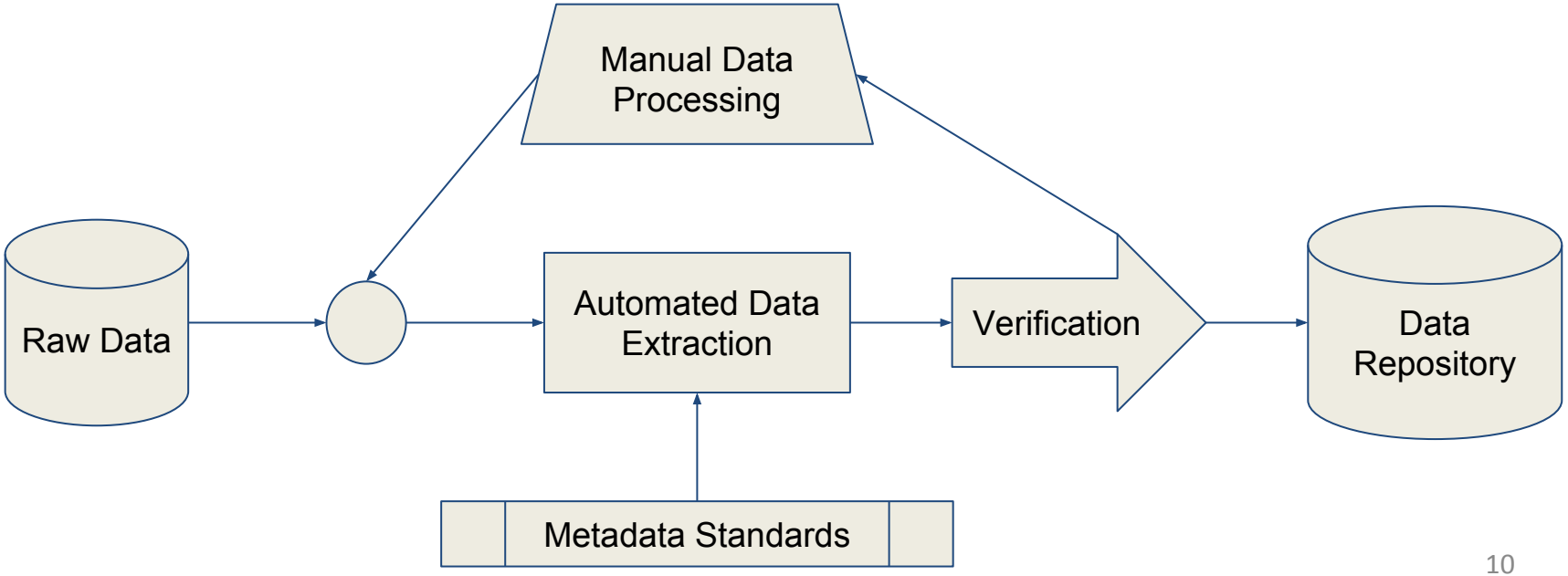
Full list: http://www.dcc.ac.uk/resources/metadata-standards

Best Practices for Metadata Creation

- Identify formats, terminologies and guidelines
  - **Conceptual model:** define the structure and interrelation of information and transmission format
  - **Controlled vocabularies:** lists of predefined terms that ensure consistency of use
  - **Technical standards:** ensure that units such as date and time, etc. are entered in consistent formats
  - **Minimum information reporting**: report the same essential information
- Identify data elements (dates, funders, methodologies, files)

Creating Metadata

Balancing between manual entry and automatic extraction



Final Considerations

- Data dictionary of the elements, technical standards, and controlled vocabularies
- Established metadata standard - your discipline probably already has a best practices metadata standard specific to your research needs
- Contact a metadata specialist - they are there to assist you!